

# Projected Cartan output ledger, full-frame Schur–Jensen boundary, and temporal scope

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-25 · this version issued 2026-07-25 · v1.0

## 1 Purpose and scope

R-090 left three deliberately separate obligations: the projected Cartan FAR estimate  $H_C$ , the complete signed linear–rational NEAR estimate  $H_N$ , and the cutoff/partition/revisit-uniform progressive assembly  $H_A$ . This note advances all three without identifying a local algebraic reduction with the missing form bound.

The results are as follows.

1. The exact output-projected Cartan trace admits a lossless relative-gap ledger beginning at the smooth-Littlewood–Paley safe separation  $C_0 = 5$ . The ledger gains  $2^{-2\gamma(C-5)}$  and avoids the two-shell loss introduced by a coefficient-annulus replacement.
2. Applying the existing R-087 remainders to this ledger produces an explicit cumulative  $Z^6$  majorant. The accepted predictable rare-event fixture makes that majorant grow like  $N^3$  while every available energy budget stays of order one. This is a proof-method no-go, not a Cartan counterexample.
3. In the exact scalar one-mode production map, the same rare fixture is instead paid by Cameron–Martin plus quartic scaling and its fixed-gap trace decays like  $N^{-4}$ . Every dyadic FAR gap is genuinely populated, but the gap tail is superexponentially small in the linearized scalar model. Thus saturation repairs the fixture and identifies the missing theorem as a cumulative, expectation-inside, vector paracomposition estimate.
4. Stacking all three linear Pauli–Fierz rows with the rational row gives an exact full-frame conditional Schur formula. Its same-root version has an explicit signed Jensen residual. No fixed local derivative payment can make the raw conditional endpoint universally positive; a complete local scalar fixture is strictly negative even after the linear radial row is included. This rejects automatic positivity, not the global form bound.
5. A terminal paid split gives the R-090 unpaid endpoint exactly and without a duplicated forest. It closes endpoint bookkeeping once the paid functional is declared exactly. It does not supply the overlap-stable temporal Cartan theorem needed for arbitrary progressive revisits.

The host claim remains T4. Projected CFAR, the output-ledger one-use bound, the complete signed NEAR packet, progressive assembly, REG, full OVERLAP, Nelson, an interacting measure, and Sector-A closure remain open.

## 2 Exact output-projected Cartan ledger

Retain R-090's exact conservative identity  $b_{A,j} = \nabla c_{A,j}$  and its output multiplier  $q$ . For each production row  $A$ , current root  $j$ , and output shell  $m$ , define the nonnegative exact block

$$\begin{aligned}\mathcal{E}_{A,j,m} &:= \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(j)} \mathbb{D}_j H_{A,j,A^{(j)}}\|_{\text{HS}}^2 dt \\ &= \int_0^\infty e^{-2t} \mathbb{E} \sum_{\mu,p} \lambda_{j,p} \sum_q |\pi_m(q)|^2 |q|^2 |\hat{c}_{A,j,t}(q-p)^T e_\mu|^2 dt.\end{aligned}\tag{2.1}$$

No expectation, OU average, root sum, output multiplier, or physical-shell cross term has been moved outside this definition. The controlled Cartan FAR energy is

$$S_C^{\text{ctrl}} = \frac{3}{40P} \mathcal{H}_C(C), \quad \mathcal{H}_C(C) = \sum_{A,j} \sum_{m \geq j+C} \mathcal{E}_{A,j,m}.\tag{2.2}$$

R-087 proved for the declared smooth analytic LP convention that every principal contribution lies below  $Q_{j+4}$ :

$$\mathcal{E}_{A,j,m}^{\text{principal}} = 0 \quad (m \geq j+5).\tag{2.3}$$

The value 5 is the safe smooth-LP constant. The sharper formal paraproduct value and the polynomial  $C \geq 3$  convention are not substituted here.

For  $\gamma > 0$ , introduce the exact output excess ledger

$$\mathcal{B}_\gamma^{\text{out}} := \sum_{A,j} \sum_{m \geq j+5} 2^{2\gamma(m-j-5)} \mathcal{E}_{A,j,m}.\tag{2.4}$$

### Theorem 2.1 (lossless output-gap extraction)

For every  $C \geq 5$  and every  $\gamma > 0$ ,

$$\boxed{\mathcal{H}_C(C) \leq 2^{-2\gamma(C-5)} \mathcal{B}_\gamma^{\text{out}}}.\tag{2.5}$$

*Proof.* For  $m \geq j+C$ , the weight in (2.4) is at least  $2^{2\gamma(C-5)}$ . Multiply each nonnegative block by the reciprocal and sum. Equation (2.3) ensures that no principal piece has been discarded.  $\square$

The gain is  $C-5$ , not  $C-7$ . The latter appears only after replacing the exact  $q$ -weighted trace by the two-shell-enlarged coefficient projector. Theorem 2.1 is an exact reduction; it becomes a one-use theorem only after  $\mathcal{B}_\gamma^{\text{out}}$  is controlled by the available energies.

## 3 The cumulative remainder majorant and its failure

Write

$$a_j = A^{(j)} - A^{(\ell)}, \quad u_{j,r} = w_j + A^{(\ell)} + r a_j, \quad Z_{j,r} = 1 + \|u_{j,r}\|_{C^\alpha}.\tag{3.1}$$

Applying R-087 (5.3)–(5.5) before physical-shell atomization gives, for  $m \geq j+5$ ,

$$\begin{aligned}\mathcal{E}_{A,j,m} &\lesssim \mathbb{E} \int_0^1 Z_{j,r}^6 [2^{-4\alpha m} 2^{(1-2\alpha)j} \|a_j\|_2^2 \\ &\quad + 2^{-6\alpha m} 2^{-j} \|D a_j\|_2^2 + 2^{-6\alpha m} 2^j \|a_j\|_2^2] dr.\end{aligned}\tag{3.2}$$

Consequently, when  $1/3 < \alpha < 1/2$  and  $0 < \gamma < 2\alpha$ ,

$$\mathcal{B}_\gamma^{\text{out}} \lesssim_{\alpha,\gamma,e} \mathcal{Z}_\alpha,\tag{3.3}$$

where

$$\mathcal{Z}_\alpha = \sum_j \mathbb{E} \int_0^1 Z_{j,r}^6 \left[ 2^{-(6\alpha-1)j} \|a_j\|_2^2 + 2^{-(6\alpha+1)j} \|Da_j\|_2^2 \right] dr. \quad (3.4)$$

The two geometric denominators are

$$\frac{2^{-20\alpha}}{1 - 2^{-(4\alpha-2\gamma)}}, \quad \frac{2^{-30\alpha}}{1 - 2^{-(6\alpha-2\gamma)}}. \quad (3.5)$$

At the production values

$$\alpha = \frac{2}{5}, \quad \gamma = \frac{7}{12}, \quad (3.6)$$

the margins are

$$4\alpha - 2\gamma = \frac{13}{30}, \quad 6\alpha - 2\gamma = \frac{37}{30}, \quad (3.7)$$

and (2.5) reads

$$\mathcal{H}_C(C) \leq 2^{-7(C-5)/6} \mathcal{B}_{7/12}^{\text{out}}. \quad (3.8)$$

The majorant (3.4) cannot close one-use. Let  $N = 2^k$ , choose a predictable event  $E \in \mathcal{F}_{k-1}$  of probability  $N^{-6}$ , and set

$$h_k = N^3 \mathbf{1}_{E \in N}, \quad a_k = K_k h_k = N \mathbf{1}_{E \in N}. \quad (3.9)$$

On a fixed positive-length interval of secant parameters,  $Z_{k,r} \asymp N^{1+\alpha}$ . The value and derivative summands in (3.4) both have exact scale power

$$\begin{aligned} -6 - (6\alpha - 1) + 6(1 + \alpha) + 2 &= 3, \\ -6 - (6\alpha + 1) + 6(1 + \alpha) + 4 &= 3. \end{aligned} \quad (3.10)$$

In contrast, the control, sextic, and mixed budgets all have scale power zero. Thus

$$\mathcal{Z}_\alpha \not\lesssim 1 + \mathbb{E} \sqrt{XY}. \quad (3.11)$$

This registers

$$\text{NG-2026-07-25-A13-PROJECTED-CARTAN-CUMULATIVE-Z6-MAJORANT}. \quad (3.12)$$

Because  $\mathcal{Z}_\alpha$  is only an upper bound, (3.11) refutes neither  $\mathcal{B}_\gamma^{\text{out}}$  nor  $\mathcal{H}_C$ .

## 4 Exact scalar saturation repairs the rare fixture

The production slice

$$F(x) = \frac{x^3}{1 + x^2} \quad (4.1)$$

is bounded at the derivative level and cannot be faithfully represented by the unbounded pathwise multiplier  $Z^6$ . Let

$$d = (1 + A^2)^{1/2}, \quad \rho = \frac{d - 1}{d + 1}, \quad u = \frac{2A}{d(d + 1)}. \quad (4.2)$$

The elementary generating function gives

$$\frac{A \cos \theta}{1 + A^2 \cos^2 \theta} = u \sum_{k=0}^{\infty} (-\rho)^k \cos((2k + 1)\theta). \quad (4.3)$$

Indeed, the series equals

$$\frac{u(1+\rho)\cos\theta}{(1-\rho)^2+4\rho\cos^2\theta} = \frac{A\cos\theta}{1+A^2\cos^2\theta}. \quad (4.4)$$

Since  $F(A\cos\theta) = A\cos\theta$  minus (4.3), every nonlinear odd harmonic is explicit. For a carrier  $N$  and  $K \geq 1$ , its exact derivative tail is

$$\frac{N^2 u^2}{2} \sum_{k \geq K} (2k+1)^2 \rho^{2k}. \quad (4.5)$$

With  $x = \rho^2$ , the sum is

$$x^K \left\{ \frac{(2K+1)^2}{1-x} + \frac{4(2K+1)x}{(1-x)^2} + \frac{4x(1+x)}{(1-x)^3} \right\}. \quad (4.6)$$

#### Lemma 4.1 (one-mode energy scaling)

For every  $s \geq 0$ ,

$$(4.5) \lesssim_s N^2 (2K+1)^{-2s} (A^2 + A^{1+2s}). \quad (4.7)$$

After the order-four root value mass  $N^{-1}$  is inserted, and for  $1/2 \leq s \leq 3/2$ ,

$$\boxed{\text{scalar trace tail} \lesssim_s (2K+1)^{-2s} (A^2 N^4 + A^4 N^2)}. \quad (4.8)$$

*Proof.* Insert  $(2k+1)^{-2s}(2k+1)^{2s}$  in (4.5). The identities  $1-\rho = 2/(d+1)$  and  $u = 2A/[d(d+1)]$ , followed by the standard geometric moment bound, give

$$u^2 \sum_{k \geq 0} (2k+1)^{2+2s} \rho^{2k} \lesssim_s A^2 + A^{1+2s}.$$

This proves (4.7). If  $A \leq 1$ ,  $A^{1+2s} \leq A^2$ ; if  $A \geq 1$  and  $s \leq 3/2$ ,  $A^{1+2s} \leq A^4$ . Since  $N \geq 1$ , multiplication by  $N^{-1}$  proves (4.8).  $\square$

For the rare fixture  $\mathbb{P}(E) = N^{-6}$  and  $A = N$ , the fixed-gap exact quantity in (4.5), including root mass and probability, is  $O(N^{-4})$ . The independent executable measures successive dyadic slopes converging to  $-4$ . Thus the  $N^3$  divergence in (3.10) is genuinely an artefact of the extracted  $Z^6$  majorant.

This scalar result is not full  $H_C$ . Arbitrary vector and multimode secants, adapted Gaussian backgrounds, target heat, root summation, and coherent output collisions remain outside Lemma 4.1.

## 5 Every gap survives, but the scalar tail is superexponential

Finite Taylor support is also unavailable. Put

$$H(\theta) = F'(\cos\theta) \cos^2\theta = \frac{\cos^4\theta(3+\cos^2\theta)}{(1+\cos^2\theta)^2}. \quad (5.1)$$

If  $y = \cos 2\theta$ , then

$$H = \frac{y}{2} + \frac{3}{2} - \frac{6}{3+y} + \frac{8}{(3+y)^2}. \quad (5.2)$$

Let

$$r = 3 - 2\sqrt{2}, \quad x = r^2 = 17 - 12\sqrt{2}. \quad (5.3)$$

Differentiating the classical Fourier series of  $(a + \cos t)^{-1}$  with respect to  $a$  gives, for every  $n \geq 2$ ,

$$\boxed{h_{2n} = (-r)^n \left( 2n - \frac{3\sqrt{2}}{2} \right)}, \quad (5.4)$$

where  $h_{2n}$  is the cosine coefficient of  $H$ . Every coefficient in (5.4) is nonzero. In particular, choosing  $n = 2^{C-1}$  produces a genuine output at the relative gap  $2^C$ .

For  $R \geq 2$ , define the linearized derivative-tail energy

$$T_R = \frac{1}{2} \sum_{n=R}^{\infty} (2nh_{2n})^2. \quad (5.5)$$

With  $S_k(R, x) = \sum_{n \geq R} n^k x^n$  and  $a = 3\sqrt{2}/2$ ,

$$T_R = 8S_4(R, x) - 8aS_3(R, x) + 2a^2S_2(R, x). \quad (5.6)$$

The shifted geometric moments give the closed forms, and the simple bound

$$\boxed{T_R \leq 8C_x R^4 x^R, \quad C_x = \frac{1 + 11x + 11x^2 + x^3}{(1-x)^5} = 1.54821660985479 \dots} \quad (5.7)$$

For  $R = 2^{C-1}$ , this is  $O(2^{4C} r^{2^C})$ , superexponential in the gap.

For the rescaled first variation,

$$T_\varepsilon[v] = \varepsilon N H'(Nx_1) + O(\varepsilon^2), \quad (5.8)$$

so (5.5) is multiplied by  $\varepsilon^2 N^2$  to leading order. Equations (5.4)–(5.8) refute exact finite-support or finite-Taylor-tree closure and the claim that FAR must live only in high current-root chaos. They do not refute a decaying FAR estimate.

If  $|p|_\infty \leq 2^j$ ,  $|r|_\infty \geq 2^{j+\Delta}$ , and  $q = r + p$ , then

$$(1 - 2^{-\Delta})^2 |r|_\infty^2 \leq |q|_2^2 \leq 3(1 + 2^{-\Delta})^2 |r|_\infty^2. \quad (5.9)$$

At  $\Delta = 5$ , the upper constant is  $3267/1024$ . This is only a coefficient-to-output comparison for the scalar diagnostic; the exact production theorem should retain  $|q|^2$  as in (2.1).

## 6 Complete linear–rational endpoint and conditional Schur form

Stack the three linear rows and rational row:

$$C(z) = \begin{pmatrix} C_L(z) \\ C_R(z) \end{pmatrix}, \quad B(z) = C(z)^T C(z) = B_L(z) + B_R(z). \quad (6.1)$$

Target-value heat acts on the Gram:

$$\bar{B}_\Sigma(z) = B_L(z) + H_\Sigma + P_\Sigma B_R(z). \quad (6.2)$$

Let

$$Z_0 = U + A^0, \quad G = DZ_0, \quad a = A^* - A^0, \quad c = Da, \quad (6.3)$$

and set

$$B_0 = \bar{B}_\Sigma(Z_0), \quad B_1 = \bar{B}_\Sigma(Z_0 + a), \quad D = B_1 - B_0. \quad (6.4)$$

Expanding the complete endpoint once gives

$$\boxed{\Delta W_{LR} = \frac{1}{2} D : (GG^T - \Gamma) + G^T B_1 c + \frac{1}{2} c^T B_1 c.} \quad (6.5)$$

The current square, Wick trace, target heat, and literal R-063 reconstruction each occur once.

Let a conditional sigma-field contain  $B_0, B_1, c, \Gamma$ , and put

$$\mu = \mathbb{E}[G \mid \mathcal{H}], \quad V = \text{Cov}(G \mid \mathcal{H}). \quad (6.6)$$

### Theorem 6.1 (conditional full-frame identity)

$$\mathbb{E}[\Delta W_{LR} \mid \mathcal{H}] = \frac{1}{2}(c + \mu)^T B_1(c + \mu) - \frac{1}{2}\mu^T B_0\mu + \frac{1}{2}(B_1 - B_0) : (V - \Gamma). \quad (6.7)$$

If  $\eta|c|^2$  is added and  $A_\eta = B_1 + 2\eta I$ , then

$$\inf_c \mathbb{E}[\Delta W_{LR} + \eta|c|^2 \mid \mathcal{H}] = \frac{1}{2}D : (V - \Gamma) + \frac{1}{2}\mu^T [2\eta B_1 A_\eta^{-1} - B_0]\mu. \quad (6.8)$$

*Proof.* Use  $\mathbb{E}[GG^T \mid \mathcal{H}] = V + \mu\mu^T$  in (6.5) and collect the quadratic terms. Completing the  $A_\eta$  square gives (6.8). The identity  $B_1 - B_1 A_\eta^{-1} B_1 = 2\eta B_1 A_\eta^{-1}$  follows by functional calculus.  $\square$

When  $V = \Gamma$ , universal conditional positivity is equivalent to

$$\begin{pmatrix} B_1 - B_0 & B_1 \\ B_1 & A_\eta \end{pmatrix} \succeq 0, \quad \text{or} \quad 2\eta B_1(B_1 + 2\eta I)^{-1} - B_0 \succeq 0. \quad (6.9)$$

But

$$2\eta B_1(B_1 + 2\eta I)^{-1} \preceq 2\eta I, \quad (6.10)$$

whereas the production  $B_0$  is unbounded. No fixed  $\eta$  makes (6.9) universal. This registers

$$\text{NG-2026-07-25-A13-FULL-FRAME-CONDITIONAL-POSITIVITY}. \quad (6.11)$$

It is a no-go for a positivity proof, not for infinitesimal form absorption.

## 7 Same-root Schur–Jensen residual

At the predictable-root conditioning actually needed by a progressive packet, the coefficients need not be measurable. Consider

$$\mathcal{E} = \frac{1}{2}D : (GG^T - \Gamma) + G^T Cc + \frac{1}{2}c^T Bc. \quad (7.1)$$

Write conditional averages with hats and define

$$\begin{aligned} r_C &= \mathbb{E}[(C - \hat{C})^T (G - \mu) \mid \mathcal{H}], \\ J_D &= \frac{1}{2}\mathbb{E}[(D - \hat{D}) : \{GG^T - (V + \mu\mu^T)\} \mid \mathcal{H}], \\ h &= \hat{C}^T \mu + r_C, \quad A_\eta = \hat{B} + 2\eta I. \end{aligned} \quad (7.2)$$

Direct conditional averaging and one square completion yield

$$\begin{aligned} \mathbb{E}[\mathcal{E} + \eta|c|^2 \mid \mathcal{H}] &= \frac{1}{2}(c + A_\eta^{-1}h)^T A_\eta(c + A_\eta^{-1}h) \\ &\quad + \frac{1}{2}\mu^T \hat{D}\mu + \frac{1}{2}\hat{D} : (V - \Gamma) + J_D - \frac{1}{2}h^T A_\eta^{-1}h. \end{aligned} \quad (7.3)$$

Thus the exact signed remainder is

$$\mathfrak{S}_{\eta, \mathcal{H}} = \frac{1}{2}\mu^T \hat{D}\mu + \frac{1}{2}\hat{D} : (V - \Gamma) + J_D - \frac{1}{2}h^T A_\eta^{-1}h. \quad (7.4)$$

The  $r_C$  and  $J_D$  terms are genuine same-root coefficient/Gaussian correlations. A terminal martingale difference of the control alone does not determine them. The missing  $H_N$  theorem is a signed Schur–Carleson estimate for the complete R-079 future-feedback block, with low and paid companions retained; an absolute factorization of the terms in (7.4) is not available.

## 8 Exact local full-frame sign fixture

On the pure-doublet scalar ray, the complete four-row coefficient is

$$b_{\text{tot}}(u) = 4c_0u^2 + 4c_1 \left( u - \frac{5}{9} \frac{u^3}{u^2 + e} \right)^2, \quad (8.1)$$

where

$$c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad P = 4 + 10^{-12}. \quad (8.2)$$

Take  $z = \sqrt{e}$ ,  $Q = G^2 - 1$ , and the branch shift

$$a = -\frac{z}{2} \quad (Q \geq 0), \quad a = +\frac{z}{2} \quad (Q < 0), \quad c = 0. \quad (8.3)$$

The exact coefficient changes are

$$\Delta b_- = -e \left( 3c_0 + \frac{35}{27}c_1 \right) < 0, \quad (8.4)$$

$$\Delta b_+ = e \left( 5c_0 + \frac{18095}{13689}c_1 \right) > 0. \quad (8.5)$$

Therefore  $\frac{1}{2}\Delta b, Q < 0$  away from  $Q = 0$ . Since  $\mathbb{E}[Q\mathbf{1}_{|G|\geq 1}] = 2\varphi(1)$  and the complementary value is  $-2\varphi(1)$ ,

$$\boxed{\mathbb{E}\Delta W_{\text{tot}} = -e\varphi(1) \left( 8c_0 + \frac{35840}{13689}c_1 \right) = -\frac{3708}{21125P}e\varphi(1) < 0.} \quad (8.6)$$

The linear radial row reinforces the rational loss; the two horizontal rows vanish on this ray.

This remains a local sign fixture. It omits the torus/A1 control realization, spatial shells,  $a = T_J v$ , global  $c = Da$ , paid subtraction, low objects, and the actual CM/sextic payments. Those payments absorb a fixed local loss. Equation (8.6) is not a post-paid  $H_N$  counterexample.

## 9 Terminal nonduplication and the temporal boundary

Let the complete endpoint be split exactly as

$$B_{\text{alg}}(H) = B_{\text{alg}}^>(H) + R_{\leq}(H). \quad (9.1)$$

R-079 gives the full-current endpoint difference

$$L_\ell + \sum_k (F_k + I_k) = \mathbb{E}[B_{\text{alg}}(A^*) - B_{\text{alg}}(A^0)]. \quad (9.2)$$

Substitution of (9.1) proves the terminal identity

$$\boxed{\begin{aligned} \mathcal{U}_{J,\pi} &:= \mathbb{E}B_{\text{alg}}^>(A^0) + L_\ell + \sum_k (F_k + I_k) \\ &\quad - \mathbb{E}[R_{\leq}(A^*) - R_{\leq}(A^0)] = \mathbb{E}B_{\text{alg}}^>(A^*). \end{aligned}} \quad (9.3)$$

This proves endpoint nonduplication for any exactly declared terminal paid functional. It also keeps  $\mathbb{E}B_{\text{alg}}^>(A^0)$  and the conditional low-current term  $L_\ell$  distinct.

For one scalar current, the indispensable future cross is visible in

$$\frac{1}{2}\{(w + f + i)^2 - w^2\} = wf + \frac{f^2}{2} + (w + f)i + \frac{i^2}{2}. \quad (9.4)$$

Replacing  $(w + f)i$  by  $wi$  loses  $fi$ . In the actual R-079 packet, the future current contains both the coefficient change on the accumulated derivative and the derivative feedback of the future control. Neither may be bounded as an unrelated addition.

Equation (9.3) does not show that the R-086 coefficient-dominant spatial classification supplies a cutoff-, partition-, and revisit-uniform estimate. The Cartan results R-084/R-087 are regular strict-past one-shot estimates, whereas full  $H_A$  quantifies over bounded cylindrical-simple progressive controls whose physical covariance ranges can overlap or coincide. The required statement is

$$\boxed{\mathbb{E}G_J(X_J + T_J v) + \frac{9}{20} \mathbb{E} \sum_k |I_k| \|u_k\|^2 \geq -C} \quad (9.5)$$

uniformly in  $J$ , time partition, and revisit multiplicity, with the complete low/future/paid packet assembled before estimation. This registers

$$\text{AUDIT-2026-07-25-A13-REG-OVERLAP-TEMPORAL-SCOPE.} \quad (9.6)$$

## 10 CORE, OVERLAP, and REG are not interchangeable

For

$$G_J = V_J^{\text{ren}} + \frac{3}{20} \|\cdot\|_6^6, \quad q = \frac{10}{9}, \quad (10.1)$$

the fixed-cutoff variational CORE theorem gives

$$-\frac{9}{10} \log \mathbb{E} e^{-(10/9)G_J(X_J)} = \inf_{v \in \mathcal{A}_{cs}} \mathbb{E} \left[ G_J(X_J + T_J v) + \frac{9}{20} \int |v|^2 \right], \quad (10.2)$$

and the same infimum over the full, bounded, or bounded cylindrical-simple classes. Hence the uniform full-OVERLAP lower bound on  $\mathcal{A}_{cs}$  is equivalent to the uniform  $q = 10/9$  Nelson estimate. No density direction is missing.

REG is useful sufficient architecture for the regular packet. It is not a hypothesis of the equivalence (10.2), and a regular one-shot bound does not imply the progressive inequality (9.5). The class inclusion and temporal overlap go in the wrong direction. The honest logical map is therefore

$$\boxed{\text{fixed-}J \text{ CORE} \quad + \quad \text{uniform full packet bound} \quad \Longleftrightarrow \quad q = 10/9 \text{ Nelson},} \quad (10.3)$$

with REG,  $H_C$ , and  $H_N$  serving as possible proof architecture rather than logically necessary labels.

## 11 Attempt, failure, and evidence map

1. **Exact output projection — proved reduction.** Keep  $q$ , OU, expectation, and the root sum together; the lossless gain is  $C - 5$ .
2. **Extracted cumulative  $Z^6$  — method falsified.** The rare fixture gives  $N^3$  with order-one budgets; do not retry this extraction.
3. **Exact scalar saturation — positive boundary.** The same fixture gives  $N^{-4}$ ; CM plus quartic scaling pays the one-mode tail.
4. **Finite Taylor support — falsified.** Every dyadic gap has a nonzero harmonic, so decay rather than exact support is required.
5. **High-chaos-only FAR — falsified as a route.** First variation reaches every gap, although its tail is superexponential.



6. **Raw conditional positivity — method falsified.** The full-frame Schur complement contains  $-B_0$  and cannot be uniformly positive for fixed  $\eta$ .
7. **Linear rows repair the rational sign — falsified locally.** The radial linear row reinforces the exact negative scalar fixture.
8. **Forest appended as repair — forbidden.** R-063 reconstructs the endpoint; the paid complement is exact subtraction.
9. **Terminal Bony/Doob split — proved algebra.** Equation (9.3) closes nonduplication, not analytic progressive estimates.
10. **REG implies OVERLAP — rejected.** General revisit/overlap needs the complete temporal theorem (9.5).

The remaining Cartan target can now be stated without the failed majorant:

$$\boxed{\mathcal{B}_{7/12}^{\text{out}} \lesssim 1 + \mathbb{E}\sqrt{XY}} \quad (11.1)$$

or a form-sufficient version with an absorbable  $\mathbb{E}X$  term. It must be proved by a saturation-aware, expectation-inside, cumulative vector paracomposition theorem. The remaining NEAR target is a signed estimate of (7.4) after the exact R-079 future/low/paid recombination. The remaining assembly target is (9.5).

## 12 Devil’s-advocate review

1. **Objection:** The output-gap ledger proves  $H_C$  because its gap factor is summable. **DISMISSED.** Summability is conditional on  $\mathcal{B}_\gamma^{\text{out}} < \infty$  with the required energy bound. That one-use estimate is exactly (11.1) and remains open.
2. **Objection:** The rare fixture disproves projected CFAR. **DISMISSED.** It diverges only after the non-equivalent  $Z^6$  majorization. The exact saturated scalar trace decreases like  $N^{-4}$ .
3. **Objection:** Scalar saturation proves the vector theorem. **VALID WITH MITIGATION.** It proves a genuine model estimate and identifies the missing saturation mechanism, but does not control coherent multimode collisions, target heat, adapted backgrounds, or the full root sum. No  $H_C$  closure is claimed.
4. **Objection:** Nonzero coefficients at every gap rule out FAR decay. **DISMISSED.** Formula (5.7) is superexponential in the dyadic gap. Nonzero support and strong decay are compatible.
5. **Objection:** The  $C_0 = 5$  separation is unnecessarily weak. **VALID WITH MITIGATION.** It is the proved safe value for the declared smooth analytic LP. Sharper constants require a different multiplier theorem and are not silently imported from the polynomial case.
6. **Objection:** Adding all linear rows restores conditional positivity. **DISMISSED.** Equations (6.8)–(6.10) include the complete linear–rational Gram, and the exact local coefficient (8.6) remains negative.
7. **Objection:** The local negative fixture falsifies the desired form bound. **DISMISSED.** It has a fixed loss absorbed by CM/sextic costs and omits the post-paid global spatial packet. It falsifies only an automatic positivity proof.

8. **Objection:** The Jensen residual can be omitted after replacing coefficients by conditional means. **DISMISSED.** The independent finite-ensemble audit verifies both  $r_C$  and  $J_D$  with nonsymmetric  $C$ ; omitting either changes (7.3).
9. **Objection:** Equation (9.3) proves  $H_A$ . **DISMISSED.** It proves endpoint algebra. Uniform control under overlapping covariance ranges and repeated visits is the separate analytic statement (9.5).
10. **Objection:** REG is part of the OVERLAP–Nelson equivalence. **DISMISSED.** CORE identifies the variational classes directly. REG is a sufficient proof architecture, not a necessary logical label.

## 13 Executable verification and no-overclaim statement

The primary executable checks authority tokens, the exact output-sequence lemma, all exponent margins, the rare-event scale powers, the scalar generating function and geometric tail, exact arbitrary-gap coefficients, three-dimensional  $q$ -shift constants, scalar and matrix Schur formulas, the nonsymmetric same-root Jensen identity, exact local fractions, and terminal polarization. It passes 72 of 72 assertions.

The independent executable imports no primary code. It uses random nonnegative output arrays, high-precision Fourier and Gaussian quadrature, an exact fixed-gap rare-tail asymptotic, random finite conditional ensembles, Schur stationarity, Jensen defects, and independent terminal sampling. It passes 24 of 24 assertions. Run

```
python codes/foundations/a13_classii_projected_cartan_full_frame_temporal_boundary_verify.py
```

R-091 proves an exact output-gap reduction, diagnoses and replaces a failed scalar majorant, derives the full-frame Schur–Jensen boundary, and closes terminal nonduplication algebra. It does not prove projected Cartan one-use, complete signed NEAR, overlap-stable progressive assembly, REG, full OVERLAP, Nelson, an interacting measure, or Sector-A closure. Tier stays T4.

## Result footer

Result ID: A13-CLASSII-PROJECTED-CARTAN-FULL-FRAME-SCHUR-  
JENSEN-TEMPORAL-BOUNDARY

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ledger:               | R-091.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Precise statement:    | The exact projected Cartan energy has a lossless C-5 output-gap ledger. The extracted cumulative Z <sup>6</sup> majorant fails, while exact scalar saturation pays the same rare fixture and has a superexponential linearized gap tail. The complete linear+rational endpoint has exact conditional Schur and same-root Jensen forms. Fixed-eta positivity fails. Terminal paid subtraction is exactly nonduplicating, while overlap-stable temporal assembly remains open. |
| Scope:                | Finite cutoff and positive floor under the pinned A1/R-079/R-084/R-087/R-089/R-090 conventions. Scalar saturation is one-mode; sign loss is local and not post-paid; temporal result is endpoint algebra.                                                                                                                                                                                                                                                                    |
| Dependencies:         | A1; R-063; R-079; R-081; R-084; R-085; R-086; R-087; R-089; R-090.                                                                                                                                                                                                                                                                                                                                                                                                           |
| Evidence grade:       | ANALYTIC, EXACT, EXECUTED, BOUNDARY.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Reproduction command: | python codes/foundations/a13_classii_projected_cartan_full_frame_temporal_boundary_verify.py                                                                                                                                                                                                                                                                                                                                                                                 |
| Expected output:      | Primary 72/72; independent 24/24; integrated and aggregate counts pinned by the manifest; exit zero.                                                                                                                                                                                                                                                                                                                                                                         |
| Negative results:     | NG-2026-07-25-A13-PROJECTED-CARTAN-CUMULATIVE-Z6-MAJORANT; NG-2026-07-25-A13-FULL-FRAME-CONDITIONAL-POSITIVITY.                                                                                                                                                                                                                                                                                                                                                              |

Audit: AUDIT-2026-07-25-A13-REG-OVERLAP-TEMPORAL-SCOPE.  
Falsification gate: Failure of output-gap arithmetic, R-087 remainder exponents, scalar series/tails, Schur/Jensen identities, local fractions, nonduplication, evidence hashes, PDF, or assertion counts.  
Tier before / after: T4 / T4; no promotion.  
No-overclaim statement: No projected CFAR, output one-use, complete signed NEAR, REG, progressive H\_A, full OVERLAP, Nelson, interacting measure, removal limit, or Sector A.  
Next required action: Prove the saturation-aware vector output-ledger bound; prove the signed Schur--Carleson NEAR packet; then prove overlap-stable temporal assembly.